



SMIE 2016 - STATS 2107 Statistical Modelling & Inference II

Statistical Modelling and Inference II (The University of Adelaide)



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**Examination in School of Mathematical Sciences
Semester 2, 2016**

104843 STATS 2107 Statistical Modelling & Inference II

Official Reading Time: 10 mins
Writing Time: 120 mins
Total Duration: 130 mins

NUMBER OF QUESTIONS: 6 TOTAL MARKS: 70

Instructions

- Attempt all questions.
- Begin each answer on a new page.
- Examination materials must not be removed from the examination room.

Materials

- 1 Blue book is provided.
- Calculators without remote communications or CAS capability are allowed.
- English and foreign-language dictionaries may be used.
- Two appendices are attached (the second appendix is a sheet of probability distributions).

DO NOT COMMENCE WRITING UNTIL INSTRUCTED TO DO SO.

1. Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed (*i.i.d.*) random variables with probability density function $f(y; \theta)$ for a real scalar parameter $\theta \in \Theta$, where Θ denotes the parameter space. Let $T = T(Y_1, Y_2, \dots, Y_n)$ be an estimator for θ .

(a) Define the mean squared error, $MSE_T(\theta)$, of T .

(b) Define the bias, $b_T(\theta)$, of T .

(c) Prove that

$$MSE_T(\theta) = \text{var}(T) + b_T(\theta)^2.$$

(d) Suppose $Y_1, Y_2, \dots, Y_n$ are *i.i.d.* Poisson random variables with rate $\lambda > 0$, and that $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ is to be used as an estimator for λ .

(i) Show that $\bar{Y}$ is an unbiased estimator for λ .

(ii) Calculate $MSE_{\bar{Y}}(\lambda)$.

[12 marks]

2. Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed $N(\mu, \sigma^2)$ random variables with σ^2 known.

(a) Carefully define a $100(1 - \alpha)\%$ confidence interval for μ .

(b) Define what is meant by a *pivotal quantity*.

(c) Suppose we want to conduct a test of $H_0 : \mu = \mu_0$ versus the two-sided alternative $H_A : \mu \neq \mu_0$ at significance level α . Assume $\mu \neq \mu_0$ and let

$$Z^* = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}.$$

(i) Show that Z^* has mean $\frac{\mu - \mu_0}{\sigma/\sqrt{n}}$ and variance 1, and hence give the distribution of Z^* .

(ii) ~~Hence show that the power function~~

~~$$P(|Z^*| \geq z(\alpha/2)) = P\left(Z \leq -z(\alpha/2) - \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right) + 1 - P\left(Z \leq z(\alpha/2) - \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right).$$~~

[12 marks]

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3. Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ are independent with $\epsilon_i \sim N(0, \sigma^2)$ for $i = 1, 2, \dots, n$. [In what follows, you may assume that

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{var}(\hat{\beta}_0) = \sigma^2 \left\{ \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right\}, \quad \text{and} \quad \text{cov}(\hat{\beta}_1, \hat{\beta}_0) = -\sigma^2 \frac{\bar{x}}{S_{xx}},$$

where $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$.]

(a) Find $\text{var}(\hat{\beta}_0 + \hat{\beta}_1 x_i)$.

(b) The i^{th} residual is defined by

$$\hat{E}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i).$$

(i) Show that $E(\hat{E}_i) = 0$.

(ii) Assuming the result

$$\text{cov}(Y_i, \hat{\beta}_0 + \hat{\beta}_1 x_i) = \sigma^2 \left\{ \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \right\},$$

$$\text{prove that } \text{var}(\hat{E}_i) = \sigma^2 \left\{ 1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{S_{xx}} \right\}.$$

[8 marks]

4. Consider the multiple regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\mathbf{Y}$ is an $n \times 1$ vector of response random variables, $\mathbf{X}$ is an $n \times p$ design matrix, $\boldsymbol{\beta}$ is a $p \times 1$ vector of regression parameters and $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of random errors with $\epsilon_i \sim N(0, \sigma^2)$, $i = 1, \dots, n$. Let $\boldsymbol{\lambda}^T \boldsymbol{\beta}$ be a linear combination of interest to be estimated.

(a) State the formula for the least squares estimate $\hat{\boldsymbol{\beta}}$.

(b) Suppose we want to make predictions for an independent random variable Y_0 at $\boldsymbol{\lambda} = \mathbf{x}_0$, where

$$Y_0 \sim N(\beta_0 + \beta_1 x_{01} + \dots + \beta_r x_{0r}, \sigma^2),$$

and $p = r + 1$.

(i) State the best linear unbiased estimate of $\boldsymbol{\lambda}^T \boldsymbol{\beta} = E(Y_0)$.

(ii) State the distribution of the prediction error $Y_0 - \hat{Y}_0$.

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(iii) Assuming σ^2 is known, show that

$$\left(\hat{Y}_0 - z(\alpha/2)\sigma\sqrt{1 + \boldsymbol{\lambda}^T(X^T X)^{-1}\boldsymbol{\lambda}}, \hat{Y}_0 + z(\alpha/2)\sigma\sqrt{1 + \boldsymbol{\lambda}^T(X^T X)^{-1}\boldsymbol{\lambda}} \right)$$

is a $100(1 - \alpha)\%$ prediction interval for Y_0 .

[9 marks]

5. A study was conducted on a sample of university students in order to determine whether a person's brain size or body size is predictive of his or her intelligence. The response variable, performance IQ score (PIQ) was measured together with the predictor variables brain size (based on the count obtained from MRI scans given as count/10,000), height in inches and weight in pounds. A multiple regression model was fitted to the data using R and the following summary output obtained (*note that two of the entries in the R output are missing and you are asked to complete the entries in the questions below*).

Call:

```
lm(formula = PIQ ~ Brain + Height + Weight, data = piq)
```

Residuals:

Min	1Q	Median	3Q	Max
-32.74	-12.09	-3.84	14.17	51.69

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	?????????	6.297e+01	1.768	0.085979 .
Brain	2.060e+00	5.634e-01	?????	0.000856 ***
Height	-2.732e+00	1.229e+00	-2.222	0.033034 *
Weight	5.599e-04	1.971e-01	0.003	0.997750

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.79 on 34 degrees of freedom

Multiple R-squared: 0.2949, Adjusted R-squared: 0.2327

F-statistic: 4.741 on 3 and 34 DF, p-value: 0.007215

- What is the value of the t -statistic for $\hat{\beta}_1$?
- How many observations are in the dataset? Justify your answer.
- Test the hypotheses

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \quad \text{vs.} \quad H_A : \text{at least one } \beta_i \neq 0, \quad i = 1, 2, 3,$$

as follows:

- State the meaning of the null hypothesis in words.

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- (ii) Write down the value of the appropriate test statistic, the associated degrees of freedom and P-value, and give your conclusion.
- (d) Do you reject or retain the null hypothesis $H_0 : \beta_2 = 0$ versus $H_A : \beta_2 \neq 0$ at the 5% significance level? Justify your answer.
- (e) What is the estimate of the intercept, β_0 ?
- (f) Assuming the critical value $t(0.025) = 2.00$, calculate a two-sided 95% confidence interval for β_1 , taking into account the other predictor variables.
- (g) Consider the diagnostic plots from the regression model fitted to the data given in Figures 1 and 2 in **Appendix A**. Are the assumptions of the model satisfied for these data? [In your answer, state clearly the assumptions of the model and whether or not each assumption is satisfied.]

[16 marks]

6. Let $Y_1, Y_2, \dots, Y_n$ be *i.i.d.* random variables with density function $f(y; \theta)$, where $\theta \in \Theta$ is a scalar parameter of interest.
- (a) Define the log-likelihood $\ell(\theta; \mathbf{y})$, the score function $S(\theta; \mathbf{y})$, and the Fisher information $i_n(\theta)$.
 - (b) Let $\hat{\theta}$ be the maximum likelihood estimator (MLE) of θ , and let θ^* be the true value of the parameter. State, without proof, the asymptotic mean, the asymptotic variance and the asymptotic distribution of $\hat{\theta}$.
 - (c) Suppose $y_1, y_2, \dots, y_n$ are independent geometric observations with parameter π , $0 < \pi < 1$. That is, for $i = 1, 2, \dots, n$,

$$f(y_i; \pi) = \pi(1 - \pi)^{y_i - 1}, \quad y_i = 1, 2, 3, \dots$$

- (i) Write down the log-likelihood $\ell(\pi; \mathbf{y})$ and find the MLE, $\hat{\pi}$, of π .
- (ii) Find the Fisher information $i_n(\pi)$.
- (iii) Consider the transformed parameter $\phi = \log \frac{\pi}{1-\pi}$. Find the maximum likelihood estimate $\hat{\phi}$.

[13 marks]

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Appendix A: Residual plots for Question 5

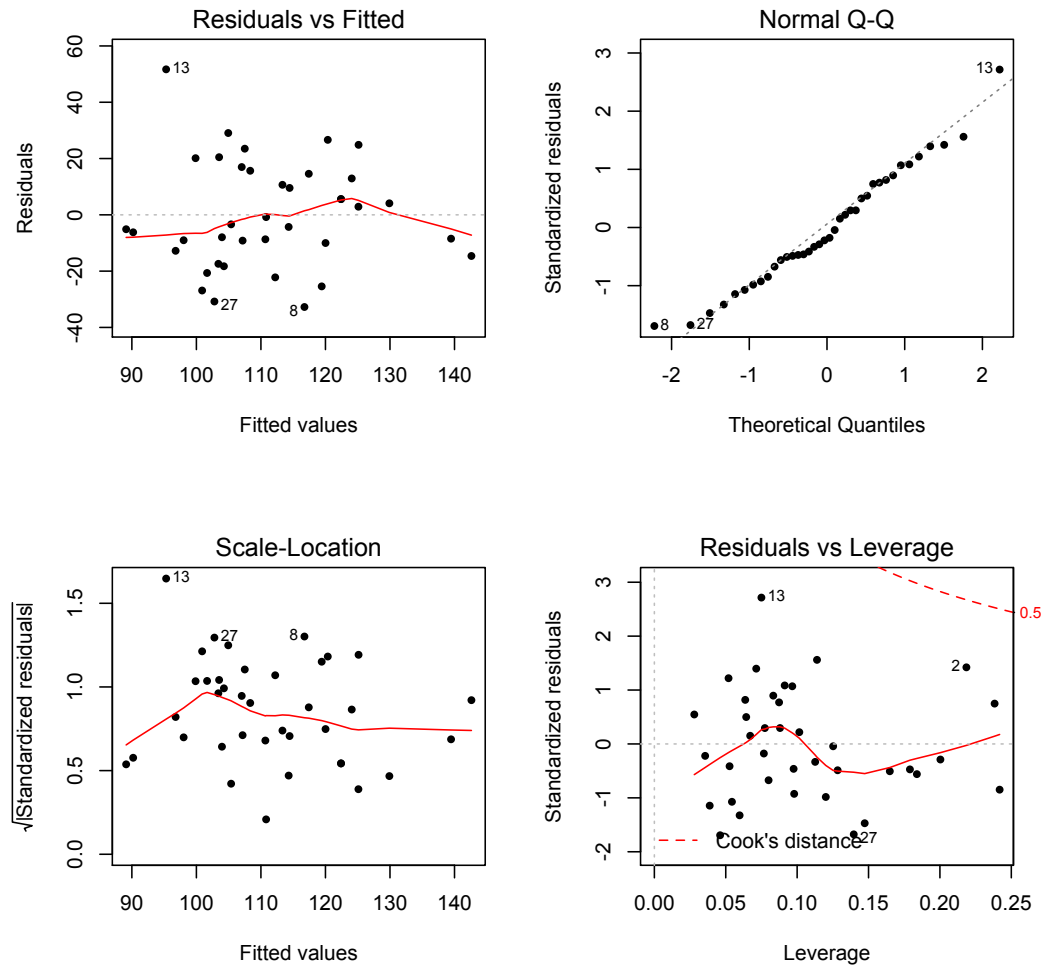


Figure 1: Diagnostic plots for Question 5.

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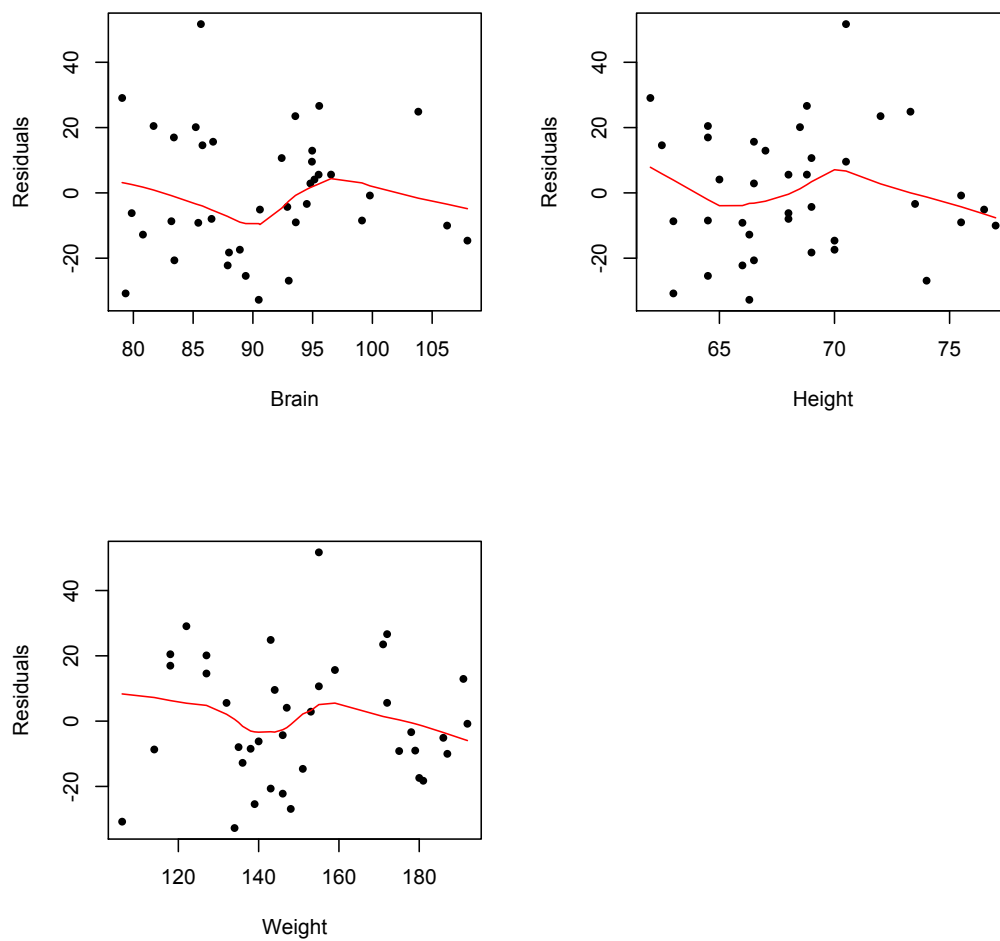


Figure 2: Residuals versus predictor variable plots for Question 5.

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Appendix B

Binomial Distribution

- $p(x) = \binom{n}{x} p^x (1-p)^{n-x}$ for $x = 0, 1, 2, \dots, n$
- $E(X) = np$
- $\text{var}(X) = np(1-p)$

Geometric Distribution

- $p(x) = p(1-p)^{x-1}$ for $x = 1, 2, \dots$
- $E(X) = \frac{1}{p}$
- $\text{var}(X) = \frac{1-p}{p^2}$

Poisson Distribution

- $p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$
- $E(X) = \lambda$
- $\text{var}(X) = \lambda$

Uniform Distribution

- $f(x) = \frac{1}{b-a}$ for $a < x < b$
- $E(X) = \frac{a+b}{2}$
- $\text{var}(X) = \frac{(b-a)^2}{12}$

Exponential Distribution

- $f(x) = \lambda e^{-\lambda x}$ for $x > 0$
- $E(X) = \frac{1}{\lambda}$
- $\text{var}(X) = \frac{1}{\lambda^2}$

Gamma Distribution

- $f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$ for $x > 0$
- $E(X) = \frac{\alpha}{\lambda}$
- $\text{var}(X) = \frac{\alpha}{\lambda^2}$

Normal Distribution

- $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1/2\sigma^2)(x-\mu)^2}$ for $-\infty < x < \infty$
- $E(X) = \mu$
- $\text{var}(X) = \sigma^2$
- $M_X(t) = e^{\mu t} e^{\sigma^2 t^2 / 2}$